

30S Series — Metal Oxide Varistors

30 mm square, epoxy-encapsulated MOVs for surge protective devices · Datasheet · Rev. 2026-10-21

Description

The 30S series is a line of 30 mm square, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. Rated In 20 kA / Imax 40 kA (8/20 µs), the standard range covers varistor voltages from 100 V to 1,800 V in 31 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.



Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- 30 mm square MOVs, epoxy encapsulated, for SPD manufacturers and OEMs
- Wide range — varistor voltage from 100 V to 1,800 V in 31 model grades
- Five grades UL 1449 Recognized, File E335803, plus DC-only recognized grades
- In 20 kA, Imax 40 kA (8/20 µs)
- Low-residual-voltage range available for tighter clamping
- Standard 30 mm square footprint used across the SPD industry

Agency approvals

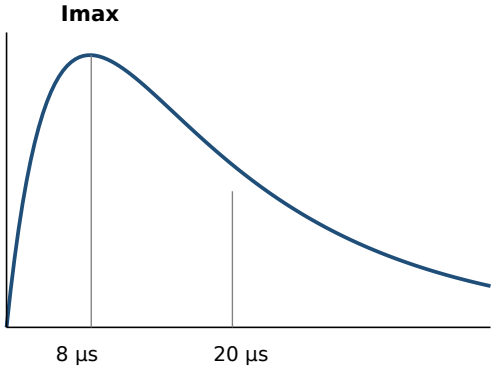
Standard	Category	File / status
UL 1449	VZCA2 (US) — surge protective components	File E335803, UL Recognized Component
CSA C22.2 No. 269.5-17	VZCA8 (Canada)	File E335803, UL Recognized Component

Five AC grades are recognized — 30SK621, 30SK681, 30SK821, 30SK112 and 30SK122 — together with the DC-only recognized grades 30SK102, 30SK751 and 30SK911. Verify in UL Product iQ (productiq.ulprospector.com) by searching file number E335803 — the record lists exactly which model codes are covered.

Absolute maximum ratings

Parameter	Range / limit	Unit
Continuous operating voltage, AC (VM(AC)rms)	60 to 1000	V
Continuous operating voltage, DC (VM(DC))	85 to 1465	V
Nominal discharge current In (8/20 µs)	20	kA
Maximum discharge current Imax (8/20 µs)	up to 40	kA
Maximum energy absorption (2 ms)	up to 1580	J

Impulse current waveform 8/20 µs (definition)



CAUTION: stresses above the ratings in the tables below may cause permanent damage. The ranges above summarize the per-model ratings in the sections below; always design to the row for the exact model code.

Model naming and standard marking

Model naming: 30S K 431 — 30 = nominal size (mm), S = square; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default). Standard marking: the manufacturer logo and the model code are etched on the encapsulation.

1. Electrical characteristics — standard ratings

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
30SK101	100	±10	60	85	20	40	360-3.6	150
30SK121	120	±10	75	100	20	40	400-3.6	185
30SK151	150	±10	95	125	20	40	450-3.0	230
30SK181	180	±10	115	150	20	40	490-2.8	270
30SK201	200	±10	130	170	20	40	520-2.6	310
30SK221	220	±10	140	180	20	40	550-2.6	350
30SK241	240	±10	150	200	20	40	600-2.6	370
30SK271	270	±10	175	225	20	40	680-2.6	430
30SK301	300	±10	195	250	20	40	770-2.6	480
30SK331	330	±10	210	270	20	40	830-2.55	520
30SK361	360	±10	230	300	20	40	910-2.55	580
30SK391	390	±10	250	320	20	40	1050-2.55	640
30SK431	430	±10	275	350	20	40	1150-2.55	730
30SK471	470	±10	300	385	20	40	1250-2.55	780
30SK511	510	±10	320	410	20	40	1300-2.55	840
30SK561	560	±10	350	450	20	40	1400-2.5	910
30SK621	620	±10	385	505	20	40	1550-2.5	990
30SK681	680	±10	420	560	20	40	1700-2.5	1000
30SK711	710	±10	440	585	20	40	1750-2.45	1010
30SK751	750	±10	460	615	20	40	1800-2.45	1020
30SK781	780	±10	485	640	20	40	1950-2.45	1030
30SK821	820	±10	510	670	20	40	2000-2.40	1080
30SK911	910	±10	550	745	20	40	2100-2.3	1170
30SK951	950	±10	575	765	20	40	2250-2.3	1180
30SK102	1000	±10	625	825	20	40	2300-2.3	1220
30SK112	1100	±10	680	895	20	40	2530-2.3	1350
30SK122	1200	±10	750	1060	20	40	2750-2.3	1400

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
30SK142	1400	±10	820	1120	20	40	3200-2.3	1450
30SK152	1500	±10	850	1240	20	40	3400-2.3	1500
30SK162	1600	±10	910	1320	20	40	3600-2.3	1540
30SK182	1800	±10	1000	1465	20	40	4000-2.3	1580

2. Electrical characteristics — low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
30SK431	430	±10	275	350	20	40	950-2.2	730
30SK471	470	±10	300	385	20	40	1030-2.2	780
30SK511	510	±10	320	410	20	40	1130-2.2	840
30SK561	560	±10	350	450	20	40	1240-2.2	910
30SK621	620	±10	385	505	20	40	1350-2.15	990
30SK681	680	±10	420	560	20	40	1480-2.15	1000
30SK711	710	±10	440	585	20	40	1540-2.15	1010
30SK751	750	±10	460	615	20	40	1630-2.15	1020
30SK781	780	±10	485	640	20	40	1670-2.15	1030
30SK821	820	±10	510	670	20	40	1850-2.1	1080
30SK911	910	±10	550	745	20	40	1940-2.1	1170
30SK951	950	±10	575	765	20	40	2040-2.1	1180
30SK102	1000	±10	625	825	20	40	2200-2.1	1220
30SK112	1100	±10	680	895	20	40	2380-2.1	1350
30SK122	1200	±10	750	1060	20	40	2620-2.1	1400

Dimensions and thickness

Encapsulated thickness Tmax (mm), standard & low-clamping ranges — 101: 3.8 · 121: 3.8 · 151: 3.8 · 181: 3.9 · 201: 4.0 · 221: 4.1 · 241: 4.2 · 271: 4.4 · 301: 4.6 · 331: 4.7 · 361: 4.9 · 391: 5.1 · 431: 5.2 · 471: 5.5 · 511: 5.7 · 561: 6.0 · 621: 6.3 · 681: 6.3 · 711: 6.6 · 751: 7.1 · 781: 7.3 · 821: 7.5 · 911: 8.1 · 951: 8.1 · 102: 8.1 · 112: 8.9 · 122: 8.9 · 142: 9.9 · 152: 10.4 · 162: 10.9 · 182: 11.9.
Dimension drawings for each package form are available on request.

Ordering and support: samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC, Bellevue, WA (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com). Specifications are subject to change without notice; verify critical ratings against the model row above before design freeze.